| | |
|---|---|
| **EQUIPMENT NAME:** | **Refer to table below** |
| **EQUIPMENT NUMBER:** | **Refer to table below** |

| **Procurement Package** | "EL008" - 415V TSF Switchboard |
|---|---|
| **Equipment No.** | **Equipment Name** |
| F75-SW-560 | FIM II TSF Cell G Switchboard 1 |
| F75-SW-593 | FIM II TSF Cell G Switchboard 2 |
| F75-SW-556 | FIM III TSF Switchboard 1 |
| F75-SW-557 | FIM III TSF Switchboard 2 |
| F75-SW-801 | FIM III TSF Switchboard 3 |
| F75-SW-802 | FIM III TSF Switchboard 4 |
| F75-SW-803 | FIM III TSF Switchboard 5 |
| F75-SW-804 | FIM III TSF Switchboard 6 |

| **Reference Documents** | | |
|---|---|---|
| **Equipment Lists** | | |
| **Document Number** | **Rev** | **Description** |
| 2000-F75-LST-EL-40001 | C | FIM II & III TSF Electrical Load List |
| 2000-F75-LST-EL-40002 | 0 | FIM II & III TSF Electrical Equipment List |
| 2000-F75-LST-EL-40003 | C | FIM II & III TSF Cable Schedule |
| 2000-F75-LST-IC-40003 | C | FIM II & III TSF DCS List |
| **Schematic Diagrams** | | |
| 2000-F75-SCM-EL-40118 | 0 | F75-SW-556 FIM III TSF Switchboard 1 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40119 | 0 | F75-SW-556 FIM III TSF Switchboard 1 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40120 | 0 | F75-SW-557 FIM III TSF Switchboard 2 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40121 | 0 | F75-SW-557 FIM III TSF Switchboard 2 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40122 | 0 | F75-SW-801 FIM III TSF Switchboard 3 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40123 | 0 | F75-SW-801 FIM III TSF Switchboard 3 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40124 | 0 | F75-SW-802 FIM III TSF Switchboard 4 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40125 | 0 | F75-SW-802 FIM III TSF Switchboard 4 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40126 | 0 | F75-SW-803 FIM III TSF Switchboard 5 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40127 | 0 | F75-SW-803 FIM III TSF Switchboard 5 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40128 | 0 | F75-SW-804 FIM III TSF Switchboard 6 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40129 | 0 | F75-SW-804 FIM III TSF Switchboard 6 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40214 | 0 | F75-SW-560 FIM II TSF Cell G Switchboard 1 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40215 | 0 | F75-SW-560 FIM II TSF Cell G Switchboard 1 Schematic Diagram Sheet 2 |
| 2000-F75-SCM-EL-40216 | 0 | F75-SW-593 FIM II TSF Cell G Switchboard 2 Schematic Diagram Sheet 1 |
| 2000-F75-SCM-EL-40217 | 0 | F75-SW-593 FIM II TSF Cell G Switchboard 2 Schematic Diagram Sheet 2 |
| **General Arrangement Drawings** | | |
| 2000-F75-DRG-EL-40111 | B | F75-SW-556 FIM III TSF Switchboard 1 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40112 | B | F75-SW-556 FIM III TSF Switchboard 1 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40113 | C | F75-SW-557 FIM III TSF Switchboard 2 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40114 | C | F75-SW-557 FIM III TSF Switchboard 2 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40115 | B | F75-SW-801 FIM III TSF Switchboard 3 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40116 | B | F75-SW-801 FIM III TSF Switchboard 3 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40117 | B | F75-SW-802 FIM III TSF Switchboard 4 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40118 | B | F75-SW-802 FIM III TSF Switchboard 4 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40119 | B | F75-SW-803 FIM III TSF Switchboard 5 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40120 | B | F75-SW-803 FIM III TSF Switchboard 5 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40121 | B | F75-SW-804 FIM III TSF Switchboard 6 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40122 | B | F75-SW-804 FIM III TSF Switchboard 6 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40211 | C | F75-SW-560 FIM II TSF Cell G Switchboard 1 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40212 | C | F75-SW-560 FIM II TSF Cell G Switchboard 1 General Arrangement Sheet 2 |
| 2000-F75-DRG-EL-40213 | C | F75-SW-593 FIM II TSF Cell G Switchboard 2 General Arrangement Sheet 1 |
| 2000-F75-DRG-EL-40214 | C | F75-SW-593 FIM II TSF Cell G Switchboard 2 General Arrangement Sheet 2 |

| **Reference Specifications** |
|---|
| 2000-F00-STS-EL-10014_2 - Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification |
| 2000-F00-STS-EL-10016_0 - Low Voltage Switchboards & Switchgear Electrical Specification |
| 2000-F00-STS-GE-10002_0 - Site Specific Information |
| 2000-F00-STS-GE-10003_2 - Plant Numbering Specification |
| 2000-F00-STS-ST-10001_2 - Steelwork Fabrication |
| 2000-F00-STS-ST-10002_2 - Steelwork Erection Specification |
| SPEC-00-G-05_12 – Technical Specification for Protective Coatings |

**Scope of Work**

**Work Included**

- Design, supply and fabrication of eight (8) outdoor switchboards including, but not limited to:
  - IP65 160A, Form 1
  - 2mm SS316, free-standing enclosure with sloping hood,
  - Incomer MCCB;
  - 250A 48-way chassis;
  - All internal Switchboard Components as specified in respective switchboard general arrangement;
  - CAT6 wiring to telemetry unit (Antenna, data radio, radio modem, relays, etc);
  - All internal wiring and cabling including power, control, copper communications and fibre communications;
  - Labelling, including stainless steel equipment nameplate;
  - Galvanised steel frame & cable way and mounting box with fixings for bolting to concrete plinth
  - Lifting lugs, as applicable;
  - Any other items which are normally supplied with this equipment;
- Surface Protection;
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment;
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly;
- Factory Acceptance Testing at manufacturer's premises;
- Supply of all necessary documentation, drawings and test documentation;

**Work Excluded**

- Cabling external to switchboard and transformer;
- Spare parts;
- Installation and commissioning at site

| Description | Unit | Duty | Vendor | Notes | Rev |
|---|---|---|---|---|---|
| **General** | | | | | |
| Standard Specification | | 2000-F00-STS-EL-10008 | | 1 | A |
| Type Tested | | Yes, provide certificates | | 1 | A |
| Site Conditions | | Refer to 2000-F00-STS-GE-10002 – Site Specific Information | | 1 | A |
| **Electrical System** | | | | | |
| Earthing System | | TN-S solidly earthed | | 1 | A |
| Nominal Voltage | V | 415 / 240 (+/- 6%) | | 1 | A |
| Phases | | 3+N+E | | 1 | A |
| Rated Frequency | Hz | 50 (+/- 2%) | | 1 | A |
| **Environment Conditions** | | | | | |
| Plant Location | | Western Australia | | 1 | A |
| Service Location | Indoor/ Outdoor | Outdoor | | 1 | A |
| Site Data | | Refer to 2000-F00-STS-GE-10002 – Site Specific Information | | 1 | A |
| **Panel Construction** | | | | | |
| Applicable Standard | | AS/NZS 61439 | | 1 | A |
| Enclosure Assembly | | Metal clad, double skinned, single-sided, sloping hood | | 1 | B |
| Ingress Protection | | IP65 | | 1 | A |
| Mounting | | Mounted to a cable way and mounting box | | 1 | B |
| Connection | | Front connected | | 1 | A |
| Form of Internal Separation | | 1 | | 1 | A |
| Original Manufacturer (AS61439.1) | | | | 2 | A |
| Assembly Manufacturer (AS61439.1) | | | | 2 | A |
| Incomer Location | | - | | 2 | B |
| Extendable Board and Bus | | Not Required | | 1 | A |
| Spare Space | | At least 20% of chassis space | | 1 | A |
| Cable Entry | | Bottom entry | | 1 | A |
| Current Rating Main Bus (3ph + N) | A | 250 | | 1 | A |
| Fault Current Rating | kA/sec | 10kA/1sec | | 1 | C |
| Current Rating Earth Bar | A | | | 2 | A |
| Busbar Configuration | | Single chassis | | 1 | A |
| Insulated Busbars | | Yes | | 1 | A |
| **Cable Way and Mounting Box** | | | | | |
| Assembly | | Galvanised steel frame | | 2 | B |
| Cable Entry | | Bottom entry from cable pit below | | 2 | B |

| Description | Unit | Duty | Vendor | Notes | Rev |
|---|---|---|---|---|---|
| Access | | Access panels to allow operational access | | 2 | B |
| Design | | Provide fixings for bolting to concrete plinth. Cable way and mounting box shall provide switchboard stability in outdoor environment. | | 2 | B |
| **Enclosure** | | | | | |
| Length - overall | mm | 1500 | | 3 | C |
| Depth – overall | mm | 500 | | 3 | A |
| Height – minimum internal | mm | | | 2 | A |
| Height – maximum external | mm | 1350 | | 3 | A |
| Height – with steel mounting box | mm | 1850 | | 3 | A |
| Mass, including equipment | kg | | | 2 | A |
| Surface protection specification, colour | | As per SPEC-00-G-05 – Technical Specification for Protective Coatings | | 1 | A |
| Escutcheon (as required) | | Gloss White | | 1 | A |
| Weather Protection | | Sloping Hood | | 2 | B |
| **Design Verification (AS 61439)** | | | | | |
| Design verification by testing | | Yes | | 1 | A |
| Design verification by comparison / reference design | | Yes - Contractor to advise | Yes | 3 | A |
| Design verification by assessment / calculation, rules | | Yes | | 1 | A |
| Statement of Compliance | | Complete Attachment A and provide documentation | | 1 | B |

| Notes |
|---|
| 1. Fixed. The Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data or compliance. |
| 2. Vendor to Advise. |
| 3. Preferred, Vendor to advise and provide reasoning for difference. |

**Vendor Drawing and Document Requirements List (VDDR)**

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

| VDDR | | | Required with Tender | Required after Award | | As Built Required |
|---|---|---|---|---|---|---|
| | | | Copies | Copies | Weeks | Copies |
| **A** | | **Production Schedule** | | | | |
| | A01 | Production Schedule / Bar Chart | | 1E | 2 | |
| | A02 | Monthly Report | | Monthly | | |
| **B** | | **Registers and General Drawing Requirements** | | | | |
| | B01 | Drawing Schedule / Register (Drawings Expected) | | 1E | 2 | |
| | B02 | General Arrangement with Foundation Loads (in AutoCAD 2023 and PDF) | | | | |
| | B03 | Cross-section | | | | |
| **C** | | **Mounting and Erection Drawings** | | | | |
| | C01 | Foundation and Mounting Details (Provided in B02) | | | | |
| | C02 | Pipe Spools | | | | |
| | C03 | Outline Dimensions, Cut-out and Support Details (Provided in B02) | | | | |
| | C04 | Match Marked Erection Drawings | | | | |
| | C05 | Motor GA (with Fixing and Shaft Details, and Overall Dimensions) | | | | |
| **D** | | **Electrical Drawings (in electronic form as appropriate)** | | | | |
| | D01 | Single Line Diagrams | | | | |
| | D02 | Schematic Drawings | | | | |
| | D03 | Terminations / Loop Diagrams | | | | |
| | D04 | Panel Layout and Fabrication Details | | 1E | 8 | 1E |
| | D05 | Nameplate | | 1E | 8 | 1E |
| | D06 | Cable Block Drawings | | | | |
| | D07 | Switchboard GAs (Overall, Incomer and Module Type) | | 1E | 8 | 1E |
| **E** | | **Process and Piping Drawings** | | | | |
| | E01 | Process Flow Sheets | | | | |
| | E02 | Process and Instrument Diagram | | | | |
| | E03 | Pneumatic or Hydraulic Diagram | | | | |
| | E04 | Lubrication Diagram | | | | |
| **F** | | **Technical Data** | | | | |
| | F01 | Equipment Information Schedule | | | | |
| | F02 | Electrical Performance data sheet | | | | |
| | F03 | Equipment List | | | | |
| | F04 | Vendor's Surface Protection Details | | 1E | 6 | |
| | F05 | Lubrication Data Schedule | | | | |
| | F06 | Material Schedule | | | | |
| | F07 | Bolt Schedule | | | | |
| | F08 | Fan Performance Curves | | | | |
| | F09 | Pump Performance Curves (with Duty Points Marked) | | | | |
| | F10 | Cable Schedule | | | | |
| | F11 | Vendor Catalogue Information | 1E | | | |
| | F12 | Calculations | | | | |
| | F13 | Completed $_{\text{Engineer}}$ Datasheets | 1E | | | |
| | F14 | Motor Datasheets | | | | |
| | F15 | Electrical Equipment List | | | | |
| | F16 | Instrument List | | | | |
| | F17 | I/O List | | | | |
| | F18 | Communications Addressing and Signal Mapping List | | | | |

| VDDR | | | Required with Tender | Required after Award | | As Built Required |
|---|---|---|---|---|---|---|
| | | | Copies | Copies | Weeks | Copies |
| **G** | | **Inspection, Certification and MDR** | | | | |
| | G01 | Inspection and Test Plans | | 1E | 6 | |
| | G02 | Inspection and Test Procedures (generic) | | | | |
| | G03 | Quality Plans and Procedures (generic) | | 1E | 4 | |
| | G04 | Manufacturer's Data Report (MDR) – Index / Contents | | 1E | 4 | |
| | G05 | Manufacturer's Data Report (MDR) | | 1E | Delivery -4 | |
| | G06 | Material Certificates | | In MDR | | |
| | G07 | Statutory Authority Certificates | | In MDR | | |
| | G08 | Test Data and Other Certificates | | In MDR | | |
| | G09 | Welding Procedure / Specification | | | | |
| | G10 | Non-Destructive Testing (NDT) Procedures | | | | |
| | G11 | Material Safety Data Sheets | | | | |
| | G12 | Commissioning Plan and Procedure | | 1E | Delivery - 4 | |
| | G13 | Structural Design Certification | | | | |
| | G14 | Performance Guarantee Statement | | | | |
| **H** | | **Shipping and Packing Details** | | | | |
| | H01 | Material / Shipping List | 1E | 1E | Delivery -4 | |
| | H02 | Package Type and Quantity (Carton, Case, Crate) | | 1E | Delivery -4 | |
| | H03 | Shipping Packages – Weights and Dimensions | | 1E | Delivery -4 | |
| | H04 | Shipping and Handling Procedure | | 1E | 12 | |
| **J** | | **Manuals** | | | | |
| | J01 | Installation, Operation and Maintenance Manual (Preliminary) | | 1E | Delivery -4 | |
| | J02 | Installation, Operation and Maintenance Manual (Final) | | 1E | Delivery | |
| **K** | | **Spare Parts Information** | | | | |
| | K01 | Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals) | 1E | 1E | 8 | |
| | K02 | Recommended Spare Parts List with Pricing (Not Included in Manuals) | 1E | 1E | Prelim 4<br>Delivery +1 | |
| | K03 | Special Tools list | 1E | 1E | 8 | |
| **M** | | **General** | | | | |
| | M01 | Control Philosophy and Logic Diagram | | | | |
| | M02 | Functional Description | | | | |

## ATTACHMENT A: VERIFICATION / STATEMENT OF COMPLIANCE WITH AS/NZS 61439.1

Supplier to complete the statement of compliance table below and provide relevant verification documentation with response.

| No. | Characteristic to be verified | Clauses[1] | Statement of Compliance | Verification type[2] | Documentation attached |
|---|---|---|---|---|---|
| 1 | Strength of material and parts: | 10.2 | | | |
| | Resistance to corrosion | 10.2.2 | | | |
| | Properties of insulating materials: | 10.2.3 | | | |
| | Thermal stability | 10.2.3.1 | | | |
| | Resistance to abnormal heat and fire due to internal electric effects | 10.2.3.2 | | | |
| | Resistance to ultra-violet (UV) radiation | 10.2.4 | | | |
| | Lifting | 10.2.5 | | | |
| | Mechanical impact | 10.2.6 | | | |
| | Marking | 10.2.7 | | | |
| 2 | Degree of protection of enclosures | 10.3 | | | |
| 3 | Clearances | 10.4 | | | |
| 4 | Creepage distances | 10.4 | | | |
| 5 | Protection against electric shock and integrity of protective circuits: | 10.5 | | | |
| | Effective continuity between the exposed conductive parts of the assembly and the protective circuit | 10.5.2 | | | |
| | Short circuit withstand strength of the protective circuit | 10.5.3 | | | |
| 6 | Incorporation of switching devices and components | 10.6 | | | |
| 7 | Internal electrical circuits and connections | 10.7 | | | |
| 8 | Terminals for external conductors | 10.8 | | | |
| 9 | Dielectric properties; | 10.9 | | | |
| | Power-frequency withstand voltage | 10.9.2 | | | |
| | Impulse withstand voltage | 10.9.3 | | | |
| 10 | Temperature rise limits | 10.10 | | | |
| 11 | Short circuit withstand strength | 10.11 | | | |
| 12 | Electromagnetic compatibility (EMC) | 10.12 | | | |
| 13 | Mechanical operation | 10.13 | | | |

[1] Refers to relevant clause or subclause in AS/NZS 61439.1.
[2] Refer to Table D.1 of AS/NZS 61439.1 for verification options.